```python
#1. Import libraries
import os
import zipfile
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from PIL import Image
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

#2. Extract the dataset
zip_path = "archive.zip"  # Just the file name if it's in the same
folder
extract_path = "rock_paper_scissors"

with zipfile.ZipFile(zip_path) as zip_ref:
    zip_ref.extractall(extract_path)

print("Dataset extracted successfully!")
```

```
Dataset extracted successfully!
```

```python
#3. Check the dataset folders
for root, dirs, files in os.walk(extract_path):
    print(root, "->", len(files), "files")
```

```
rock_paper_scissors -> 1 files
rock_paper_scissors\paper -> 712 files
rock_paper_scissors\rock -> 726 files
rock_paper_scissors\rps-cv-images -> 1 files
rock_paper_scissors\rps-cv-images\paper -> 712 files
rock_paper_scissors\rps-cv-images\rock -> 726 files
rock_paper_scissors\rps-cv-images\scissors -> 750 files
rock_paper_scissors\scissors -> 750 files
```

```python
#4. Find the actual dataset folder
for root, dirs, files in os.walk(extract_path):
    if "rock" in dirs and "paper" in dirs and "scissors" in dirs:
        dataset_path = root
        break

print("Dataset path:", dataset_path)
```

```
Dataset path: rock_paper_scissors
```

```python
#5. Check classes
classes = ["rock", "paper", "scissors"]

for cls in classes:
```

```
    path = os.path.join(dataset_path, cls)
    print(cls, ":", len(os.listdir(path)), "images")
```

```
rock : 726 images
paper : 712 images
scissors : 750 images
```

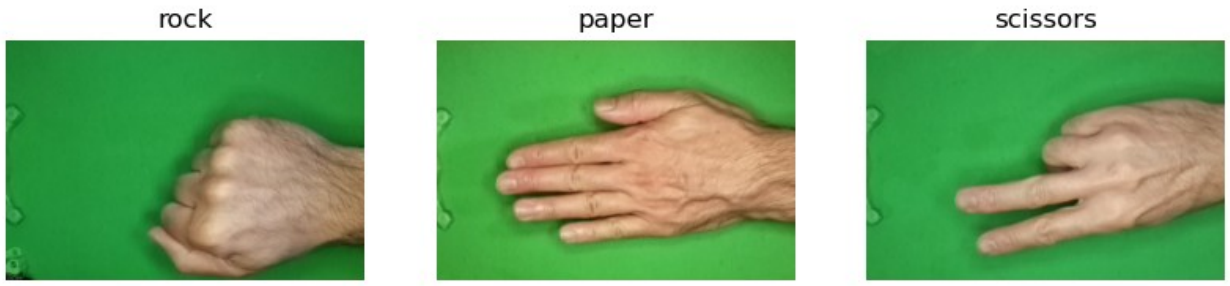
```
#6. Display sample images
plt.figure(figsize=(10, 6))

for i, cls in enumerate(classes):
    path = os.path.join(dataset_path, cls)
    image_name = os.listdir(path)[0]
    image_path = os.path.join(path, image_name)

    img = Image.open(image_path)

    plt.subplot(1, 3, i + 1)
    plt.imshow(img)
    plt.title(cls)
    plt.axis("off")

plt.show()
```




```
#7. Check image dimensions
for cls in classes:
    path = os.path.join(dataset_path, cls)
    image_name = os.listdir(path)[0]
    image_path = os.path.join(path, image_name)

    img = Image.open(image_path)

    print(cls)
    print("Image size:", img.size)
    print("Image mode:", img.mode)
    print()
```

```
rock
Image size: (300, 200)
Image mode: RGB
```

```
paper
Image size: (300, 200)
Image mode: RGB

scissors
Image size: (300, 200)
Image mode: RGB
```

```
#8. Create a basic dataset summary
data = []

for cls in classes:
    class_path = os.path.join(dataset_path, cls)

    for image_name in os.listdir(class_path):
        image_path = os.path.join(class_path, image_name)

        data.append({
            "image_path": image_path,
            "label": cls
        })

df = pd.DataFrame(data)

print("Dataset shape:", df.shape)
df.head()
```

```
Dataset shape: (2188, 2)
```

	image_path	label
0	rock_paper_scissors\rock\00nKV8oHuTGi20gq.png	rock
1	rock_paper_scissors\rock\01dM3ewPIsnBICu1.png	rock
2	rock_paper_scissors\rock\02vG75hQW9Vp4oTl.png	rock
3	rock_paper_scissors\rock\0bioBZYFCXqJIulm.png	rock
4	rock_paper_scissors\rock\0NDYNEoDui7o64gU.png	rock

```
df.describe(include="all")
```

	image_path	label
count	2188	2188
unique	2188	3
top	rock_paper_scissors\rock\00nKV8oHuTGi20gq.png	scissors
freq	1	750

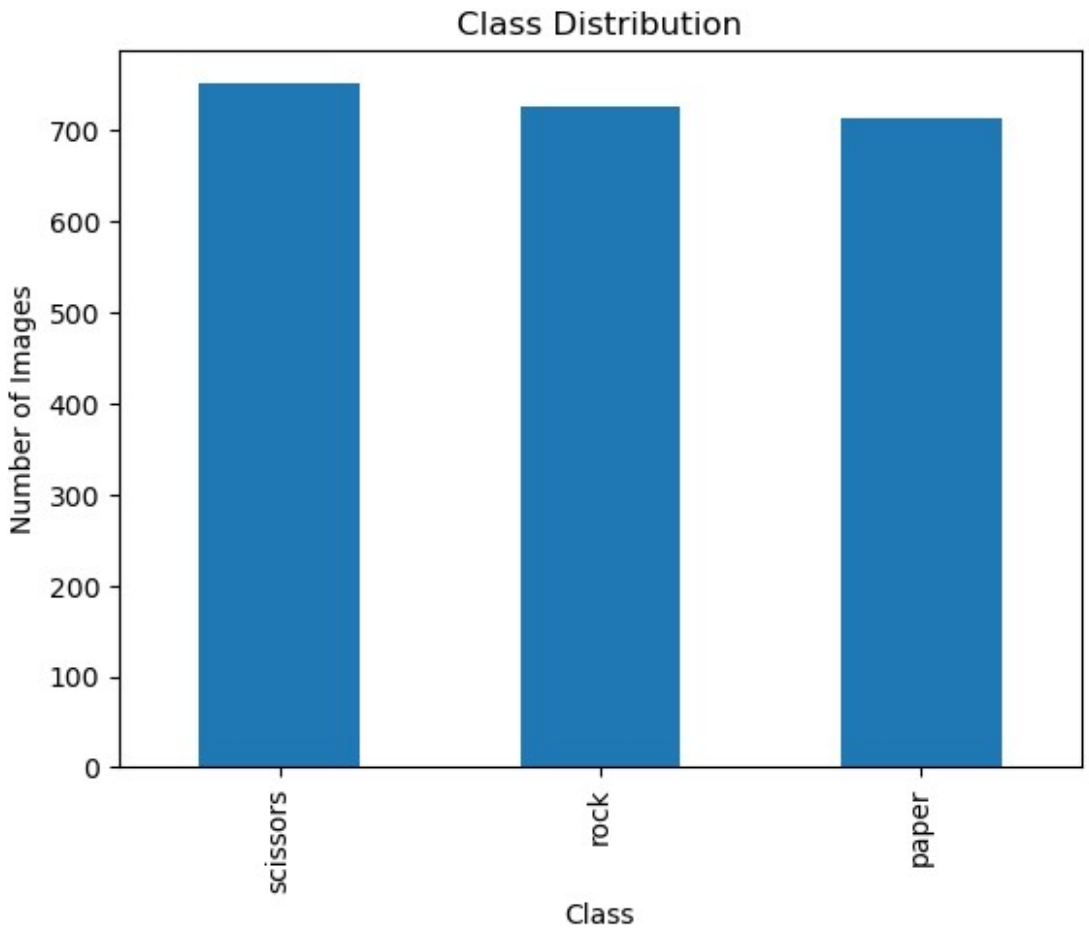
```
#9. Check class distribution
print(df["label"].value_counts())
```

```
label
scissors    750
rock        726
```

```
paper       712
Name: count, dtype: int64
```

```
df["label"].value_counts().plot(kind="bar")

plt.title("Class Distribution")
plt.xlabel("Class")
plt.ylabel("Number of Images")
plt.show()
```




```
# 10. Load your images from the extracted dataset folder
import cv2
data_dir = "rock_paper_scissors"  # Change this folder name if yours
is different

X = []
y = []

print("Loading and resizing images...")
for label_name in os.listdir(data_dir):
```

```
    class_folder = os.path.join(data_dir, label_name)
    if os.path.isdir(class_folder):
        for img_name in os.listdir(class_folder):
            img_path = os.path.join(class_folder, img_name)
            img = cv2.imread(img_path)
            if img is not None:
                img = cv2.resize(img, (64, 64))  # Resize to 64x64 for
speed
                X.append(img.flatten())          # Flatten image to 1D
array
                y.append(label_name)

X = np.array(X)
y = np.array(y)
```

```
---------------------------------------------------------------------------
-----
ModuleNotFoundError                       Traceback (most recent call
last)
Cell In[13], line 2
      1 # 10. Load your images from the extracted dataset folder
----> 2 import cv2
      3 data_dir = "rock_paper_scissors"  # Change this folder name if
yours is different
      5 X = []

ModuleNotFoundError: No module named 'cv2'
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y_encoded,
    test_size=0.20,
    random_state=42,
    stratify=y_encoded
)

print("Training data:", len(X_train))
print("Testing data:", len(X_test))

from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

lr_model = LogisticRegression(
    max_iter=300,
    solver="lbfgs",
    random_state=42
)

lr_model.fit(X_train, y_train)

y_pred_lr = lr_model.predict(X_test)
```

```
print("Logistic Regression Accuracy:",
      accuracy_score(y_test, y_pred_lr))

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred_lr,
    target_names=label_encoder.classes_
))
```